

The Dark Spooky Forest

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Graphics

4/15/19 (at the end of class)

1. Objectives

- A. Lighting, Creating an Interactive Graphical Application, Managing Complex Scenes
 - a. Make sure you know how to use Point lights
 - b. Make sure you know how to use Spot Lights
 - c. Make sure you understand how to use the directional light.

2. Equipment

- A. A computer that can run a web-browser with WebGL

3. Lab Preparation

- A. Make sure you have a strong knowledge of the lighting tutorials we went over.

4. Lab Instruction

- A. The scene
 - a. You will need three objects for this scene
 - b. A grass floor (or dirt)
 - c. A rock, must look like an obstacle and must have at least 8 sides
 - d. A tree. (It must have a spherical shape for the leaves and a box for the trunk)
 - i. Appropriately colored.
 - e. You will create a plane with trees and rocks scattered around the scene
 - i. (this can be random or placed purposefully)
 - f. The trees and the rocks must stop the camera from passing through it WITHOUT clipping through the object
- B. Ambient Light.
 - a. The ambient light will be .1
- C. Directional light.
 - a. The directional light will be coming from (20,5,20).
 - b. However, the directional light will be “moonlight” so multiply the directional light’s power by .25 to keep it pretty dim.
- D. Point lights
 - a. You will need 5 point lights (at least) scattered through the environment
 - b. They will also need to have distance fall-off. (AKA, the farther away the lower the affect)
- E. You will need a spotlight shining down and patrolling through the forest.
 - a. The position of the spotlight will be in the air pointing down.
 - b. Feel free to make a box or something that floats along with it to mark its location
 - c. If the player touches the spotlight on the ground you will be sent back to the beginning.
 - i. (you will have to figure out the radius of the spot light on the ground or a close approximation)
- F. Movement

- a. The camera should be able to move forward backward, turn left and right, and strafe left and right. (You do not need to move up and down).
 - b. Your objective is to reach a certain spot.
- G. Extra Credit:
- a. Since light does not have an object, I will give extra credit if you create
 - b. A moon object to represent the directional light.
 - i. (remember it should be really far away around 200 if you are using are 500 far-plane camera).
 - c. Candles (or torches) for all the point lights
 - d. Some sort of “bad guy” that has the spotlight.

5. Lab Rubric

	Perfect	Runs, but not quite correct	Syntax errors, but reasonable effort	Not acceptable
Rocks setup and look as though they are obstructive	10	5	2	0
Trees modeled as spheres on top of cubes	10	5	2	0
Ambient and directional light correct (directional light is dimmed)	10	5	2	0
5 Point lights	20	10	5	0
Spotlight moves and will reset the player if it touches it.	20	15	10	0
Total	/70			
Extra Credit	Moon, candles and enemy embodied (10)	Everything embodied but the moon is not far away, or the moon is not finished. (5)	Only the point lights (3)	0

6. Lab Report Requirements

There is no report due, but there is a peer review.